

CHAPTER ONE

1:

(a) Rise time(t_r):

Maximum amplitude(V_{max}) = 10 V

10% level = $0.1 \times 10V = 1V$, $t_1 = 0.1 \mu s$

90% level = $0.9 \times 10 V = 9V$, $t_2 = 0.9 \mu s$

$t_r = t_2 - t_1 = 0.9 \mu s - 0.1 \mu s = \underline{0.8 \mu s}$

(b) Fall Time (t_f):

-Edge starts falling at $t = 2.8 \mu s$ (10 V) and reaches ground at $t = 4 \mu s$ (0 V),

$4 \mu s - 2.8 \mu s = 1.2 \mu s$ (total slope duration)

To find where 90%(9V) and 10%(1V) lie on this slope

$t(v) = 2.8 + ((10-V)/10) \times 1.2$

- For 90%(9V)

$t_1 = 2.8 + ((10 - 9)/10) \times 1.2 = 2.8 + (0.1 \times 1.2) = 2.92 \mu s$

- For 10%(1V)

$t_2 = 2.8 + ((10 - 1)/10) \times 1.2 = 2.8 + (0.9 \times 1.2) = 3.88 \mu s$

$t_f = t_2 - t_1 = 3.88 \mu s - 2.92 \mu s = \underline{0.96 \mu s}$

(c) Pulse Width (t_w)

50% = $0.5 \times 10V = 5V$

Now we find where pulse crosses 5V(between 0 μs and 1 μs) and (between 2.8 μs and 4 μs)

$t_1 = 0.5 \mu s$,

$t_2 = 2.8 + ((10-5)/10) \times 1.2 = 3.4 \mu s$

$t_w = t_f = t_2 - t_1 = 3.4 \mu s - 0.5 \mu s = \underline{2.9 \mu s}$

(d) Amplitude(A) = 10V

2:

$T = 5ms - 1 ms = \underline{4ms}$

3:

$f = 1/T$

$T = 4ms = 4 \times 10^{-3} \text{ seconds}$

$= 1 / (4 \times 10^{-3} s) = \underline{250 Hz}$

4:

It's periodic because it repeats at regular intervals

5:

$$\begin{aligned}\text{Duty cycle} &= (\text{Pulse width} / \text{Period}) \times 100\% \\ &= (t_w / T) \times 100\% \\ &= (2\text{ms} / 4\text{ms}) \times 100\% = 0.5 \times 100\% = \underline{\underline{50\%}}\end{aligned}$$

6: Determining the bit sequence from a digital waveform

Given bit time = $1\mu\text{s}$

10110010

7:

$$\begin{aligned}t(\text{serial}) &= 8\text{bits} \times 1\mu\text{s} = \underline{\underline{8\mu\text{s}}} \\ t(\text{parallel}) &= \underline{\underline{1\mu\text{s}}}\end{aligned}$$

8:

$$\begin{aligned}T &= 1 / f \\ &= 1 / 3.5 \times 10^9 \text{ Hz} = \underline{\underline{2.857 \times 10^{-10}\text{s}}} = \underline{\underline{0.286 \text{ ns}}}\end{aligned}$$

9:

The light is ON if SW1 is closed or SW2 is closed, otherwise the light is off.

10: AND gate

11: AND gate

12: OR gate

CHAPTER TWO

1: Binary to decimal

a) 110011.11

Integer part (110011)

$$\begin{aligned}1 \times 2^5 &= 32 \\ 1 \times 2^4 &= 16 \\ 0 \times 2^3 &= 0 \\ 0 \times 2^2 &= 0 \\ 1 \times 2^1 &= 2 \\ 1 \times 2^0 &= 1 \quad \text{Sum} = 51\end{aligned}$$

Fractional part(.11)

$$\begin{aligned}1 \times 2^{-1} &= 1 \times 0.5 = 0.5 \\ 1 \times 2^{-2} &= 1 \times 0.25 = 0.25 \quad \text{Sum} = 0.75\end{aligned}$$

Total sum = **51.75**

b) 101010.01

Integer part (101010)

$$1 \times 2^5 = 32$$

$$0 \times 2^4 = 0$$

$$1 \times 2^3 = 8$$

$$0 \times 2^2 = 0$$

$$1 \times 2^1 = 2$$

$$0 \times 2^0 = 0 \quad \text{Sum} = 42$$

Fractional part(.01)

$$0 \times 2^{-1} = 0$$

$$1 \times 2^{-2} = 1 \times 0.25 = 0.25 \quad \text{Sum} = 0.25$$

Total sum = **42.25**

c) 1000001.111

Integer part

$$1 \times 2^6 = 64$$

$$0 \times 2^5 = 0$$

$$0 \times 2^4 = 0$$

$$0 \times 2^3 = 0$$

$$0 \times 2^2 = 0$$

$$0 \times 2^1 = 0$$

$$1 \times 2^0 = 1 \quad \text{Sum} = 65$$

Fractional part(.111)

$$1 \times 2^{-1} = 0.5$$

$$1 \times 2^{-2} = 0.25$$

$$1 \times 2^{-3} = 0.125 \quad \text{Sum} = 0.875$$

Total sum = **65.875**

d) Integer sum(1111000) = 120

Fractional sum(.101) = 0.625

Total sum = **120.625**

e) Integer sum(1011100) = 92

Fractional sum(10101) = 0.65625

Total sum = **92.65625**

f) Integer sum(1110001) = 113

Fractional sum(.0001) = 0.0625

Total sum = **113.0625**

2)

Maximum value = $2^n - 1$

- a) Two, $2^2 - 1 = \underline{3}$
- b) Three, $2^3 - 1 = \underline{7}$
- c) Four, $2^4 - 1 = \underline{15}$
- d) Five, $2^5 - 1 = \underline{31}$
- e) Six, $2^6 - 1 = \underline{63}$
- f) Seven, $2^7 - 1 = \underline{127}$
- g) Eight, $2^8 - 1 = \underline{255}$
- h) Nine, $2^9 - 1 = \underline{511}$
- i) Ten, $2^{10} - 1 = \underline{1023}$
- j) Eleven, $2^{11} - 1 = \underline{2047}$

3) Number to binary using division by 2

- a) 15 to binary
 - $15 / 2 = 7$ with remainder 1 $\rightarrow$ LSB
 - $7 / 2 = 3$ with remainder 1
 - $3 / 2 = 1$ with remainder 1
 - $1 / 2 = 0$ with remainder 1 $\rightarrow$ MSB

From bottom to top = **1111**

- b) 21 to binary
 - $21 / 2 = 10$, remainder = 1 $\rightarrow$ LSB
 - $10 / 2 = 5$, remainder = 0
 - $5 / 2 = 2$, remainder = 1
 - $2 / 2 = 1$, remainder = 0
 - $1 / 2 = 0$, remainder = 1 $\rightarrow$ MSB

From bottom to top = **10101**

- c) 28 to binary
 - $28 / 2 = 14$, remainder = 0 $\rightarrow$ LSB
 - $14 / 2 = 7$, remainder = 0
 - $7 / 2 = 3$, remainder = 1
 - $3 / 2 = 1$, remainder = 1
 - $1 / 2 = 0$, remainder = 1 $\rightarrow$ MSB

From bottom to top = **11100**

- d) 34 to binary
 - $34 / 2 = 17$, remainder = 0 $\rightarrow$ LSB
 - $17 / 2 = 8$, remainder = 1

$8 / 2 = 4$, remainder = 0
 $4 / 2 = 2$, remainder = 0
 $2 / 2 = 1$, remainder = 0
 $1 / 2 = 0$, remainder = 1 —>MSB

From bottom to top = **100010**

e) 40 to binary
 $40 / 2 = 20$, remainder = 0—>LSB
 $20 / 2 = 10$, remainder = 0
 $10 / 2 = 5$, remainder = 0
 $5 / 2 = 2$, remainder = 1
 $2 / 2 = 1$, remainder = 0
 $1 / 2 = 0$, remainder = 1—>MSB

From bottom to top = **101000**

f) 59 to binary
 $59 / 2 = 29$, remainder = 1—>LSB
 $29 / 2 = 14$, remainder = 1
 $14 / 2 = 7$, remainder = 0
 $7 / 2 = 3$, remainder = 1
 $3 / 2 = 1$, remainder = 1
 $1 / 2 = 0$, remainder = 1—>MSB

From bottom to top = **111011**

g) 65 to binary
 $65 / 2 = 32$, remainder = 1—>LSB
 $32 / 2 = 16$, remainder = 0
 $16 / 2 = 8$, remainder = 0
 $8 / 2 = 4$, remainder = 0
 $4 / 2 = 2$, remainder = 0
 $2 / 2 = 1$, remainder = 0
 $1 / 2 = 0$, remainder = 1—>MSB

From bottom to top = **1000001**

h) 73 to binary
 $73 / 2 = 36$, remainder = 1—>LSB
 $36 / 2 = 18$, remainder = 0
 $18 / 2 = 9$, remainder = 0
 $9 / 2 = 4$, remainder = 1
 $4 / 2 = 2$, remainder = 0

$$2 / 2 = 1, \text{ remainder} = 0$$

$$1 / 2 = 0, \text{ remainder} = 1 \rightarrow \text{MSB}$$

From bottom to top = **1001001**

4) Adding binary numbers

- a) $11 + 01 = \mathbf{100}$
- b) $10 + 10 = \mathbf{100}$
- c) $101 + 11 = \mathbf{1000}$
- d) $111 + 110 = \mathbf{1101}$
- e) $1001 + 101 = \mathbf{1110}$
- f) $1101 + 1011 = \mathbf{11000}$

5) Expressing decimal number as an 8-bit number in 2's complement form

- a) +12

To binary first

$$12 / 2 = 6, r = 0$$

$$6 / 2 = 3, r = 0$$

$$3 / 2 = 1, r = 1$$

$$1 / 2 = 0, r = 1$$

Binary form = 1100, Pad with leading zeros = **00001100**

- b) -68

$$68 / 2 = 34, r = 0$$

$$34 / 2 = 17, r = 0$$

$$17 / 2 = 8, r = 1$$

$$8 / 2 = 4, r = 0$$

$$4 / 2 = 2, r = 0$$

$$2 / 2 = 1, r = 0$$

$$1 / 2 = 0, r = 1$$

Binary form = 1000100, Pad with leading zeros = 01000100,
invert because negative, 10111011

Add 1 to get 2's complement = 1011101 + 1 = **10111100**

- c) +101

$$101 / 2 = 50, r = 1$$

$$50 / 2 = 25, r = 0$$

$$25 / 2 = 12, r = 1$$

$$12 / 2 = 6, r = 0$$

$$6 / 2 = 3, r = 0$$

$$3 / 2 = 1, r = 1$$

$$1 / 2 = 0, r = 1$$

Binary form = 1100101, Pad with leading zeros = **01100101**

d) -125

$$125 / 2 = 62, r = 1$$

$$62 / 2 = 31, r = 0$$

$$31 / 2 = 15, r = 1$$

$$15 / 2 = 7, r = 1$$

$$7 / 2 = 3, r = 1$$

$$3 / 2 = 1, r = 1$$

$$1 / 2 = 0, r = 1$$

Binary form = 1111101, Pad with leading zeros = 01111101,
invert because negative, 10000010

Add 1 to get 2's complement = 10000010 + 1 = **10000011**

6. Addition in 2's complement form

a) 10001100 + 00111001

$$\begin{array}{r} 10001100 \\ + 00111001 \\ \hline \end{array}$$

11000101

b) 11011001 + 11100111

$$\begin{array}{r} 11011001 \\ + 11100111 \\ \hline \end{array}$$

11000000